

From four levels to the Dicke model

a step-by-step reduction for cavity Rydberg-EIT electrometry

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Abstract

We reduce the four-level Rydberg-EIT ladder in a cavity to a collective spin–boson model, in three moves: adiabatic elimination of the intermediate state, rotation into the microwave-dressed Rydberg basis, and a Holstein–Primakoff expansion for N atoms. Each step is a matrix operation and each carries a validity condition, stated as we go. The punchline is that the microwave dressing leaves *two* coupled branches, not one, so the natural object is a two-branch Dicke model; the familiar two-level Dicke model is a limit of it, valid only when one branch is far off resonance. Throughout $\hbar = 1$.

1 The system

Four atomic levels, in the order we will keep them:

$$|1\rangle = |g\rangle = |5S_{1/2}\rangle, \quad |2\rangle = |e\rangle = |5P_{3/2}\rangle, \quad |3\rangle = |r\rangle = |nD\rangle, \quad |4\rangle = |r'\rangle = |n'P\rangle.$$

Three fields act. The *cavity mode* $\hat{a}$ drives $|1\rangle \leftrightarrow |2\rangle$ with single-atom coupling g_0 ; a classical *control* laser drives $|2\rangle \leftrightarrow |3\rangle$ with Rabi frequency Ω_c ; and the *microwave field to be measured* drives $|3\rangle \leftrightarrow |4\rangle$ with $\Omega_\mu = dE_\mu/\hbar$.

Go to the frame rotating at the drive frequencies and make the rotating-wave approximation. With Δ the one-photon detuning of the cavity from $|1\rangle \rightarrow |2\rangle$, δ the two-photon detuning of $|1\rangle \rightarrow |3\rangle$, and the microwave on resonance, the single-atom Hamiltonian in the basis $(|1\rangle, |2\rangle, |3\rangle, |4\rangle)$ is

$$H_{\text{at}} = \begin{pmatrix} 0 & g_0\hat{a}^\dagger & 0 & 0 \\ g_0\hat{a} & -\Delta & \Omega_c/2 & 0 \\ 0 & \Omega_c/2 & \delta & \Omega_\mu/2 \\ 0 & 0 & \Omega_\mu/2 & \delta \end{pmatrix}, \quad H = \omega_c \hat{a}^\dagger \hat{a} + H_{\text{at}}. \quad (1)$$

ω_c is the cavity detuning in this frame — an effective frequency, freely tunable, not an optical one. Note the structure: the matrix is *tridiagonal*, because each field connects only neighbouring levels.

2 Step 1: eliminate the intermediate state

The excited state $|2\rangle$ is short-lived and we never want it populated. Detune far from it,

$$|\Delta| \gg \Omega_c, \quad g_0\sqrt{\langle \hat{a}^\dagger \hat{a} \rangle}, \quad \Gamma_e, \quad (C1)$$

so that its amplitude follows the others adiabatically. Setting $\dot{c}_2 = 0$ in $i\dot{c}_2 = g_0\hat{a}c_1 - \Delta c_2 + (\Omega_c/2)c_3$ gives

$$c_2 \simeq \frac{1}{\Delta} \left(g_0\hat{a}c_1 + \frac{\Omega_c}{2}c_3 \right). \quad (2)$$

Substituting back and keeping the terms generated in $\{|1\rangle, |3\rangle, |4\rangle\}$:

$$H_{\text{at}}^{(3)} = \begin{pmatrix} \frac{g_0^2 \hat{a}^\dagger \hat{a}}{\Delta} & \lambda \hat{a}^\dagger & 0 \\ \lambda \hat{a} & \delta + \frac{\Omega_c^2}{4\Delta} & \Omega_\mu/2 \\ 0 & \Omega_\mu/2 & \delta \end{pmatrix}, \quad \boxed{\lambda = \frac{g_0 \Omega_c}{2\Delta}} \quad (3)$$

Three things were produced. The off-diagonal $\lambda \hat{a}$ is the *two-photon Raman coupling* $|1\rangle \leftrightarrow |3\rangle$ we want. The diagonal entries are light shifts: $\Omega_c^2/4\Delta$ on $|3\rangle$, which we absorb by redefining $\delta \rightarrow \delta + \Omega_c^2/4\Delta$, and $g_0^2 \hat{a}^\dagger \hat{a}/\Delta$ on $|1\rangle$, a photon-number-dependent (Kerr) shift which is negligible for $g_0^2 \langle \hat{a}^\dagger \hat{a} \rangle \ll |\Delta| \omega_c$ and which we drop.

Practical warning. Both λ and the light shift $\Omega_c^2/4\Delta$ scale with the control-laser intensity. The shift adds directly to δ , which is the same channel the measurand uses — so control-laser intensity noise is indistinguishable from signal. This is the dominant technical systematic of the scheme, not a detail.

3 Step 2: rotate into the dressed Rydberg basis

The lower-right 2×2 block of Eq. (3) is the microwave problem alone:

$$H_\mu = \begin{pmatrix} \delta & \Omega_\mu/2 \\ \Omega_\mu/2 & \delta \end{pmatrix} = \delta \mathbf{1} + \frac{\Omega_\mu}{2} \sigma_x. \quad (4)$$

Diagonalise it with the orthogonal rotation

$$R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad |\pm\rangle = \frac{|3\rangle \pm |4\rangle}{\sqrt{2}}, \quad R^\dagger H_\mu R = \begin{pmatrix} \omega_+ & 0 \\ 0 & \omega_- \end{pmatrix}, \quad (5)$$

$$\boxed{\omega_\pm = \delta \pm \frac{\Omega_\mu}{2}} \quad (6)$$

These are the Autler–Townes doublet: *the measurand has become a pair of level positions*. Now apply the same rotation to the whole 3×3 matrix. The only subtlety is what happens to the Raman coupling, which connects $|1\rangle$ to the *bare* $|3\rangle$:

$$|3\rangle = \frac{|+\rangle + |-\rangle}{\sqrt{2}} \implies \lambda \hat{a} |3\rangle \langle 1| = \frac{\lambda}{\sqrt{2}} \hat{a} (|+\rangle \langle 1| + |-\rangle \langle 1|). \quad (7)$$

In the basis $(|g\rangle, |+\rangle, |-\rangle)$,

$$\boxed{H_{\text{at}}^{(3)} = \begin{pmatrix} 0 & \frac{\lambda}{\sqrt{2}} \hat{a}^\dagger & \frac{\lambda}{\sqrt{2}} \hat{a}^\dagger \\ \frac{\lambda}{\sqrt{2}} \hat{a} & \omega_+ & 0 \\ \frac{\lambda}{\sqrt{2}} \hat{a} & 0 & \omega_- \end{pmatrix}} \quad (8)$$

This matrix is the heart of the reduction, and it says three things worth stating plainly:

1. **Both** dressed states couple to $|g\rangle$ — there is no basis in which only one does.

2. They couple **equally**, each with $\lambda/\sqrt{2}$. The dressing is symmetric; nothing here distinguishes $|+\rangle$ from $|-\rangle$.
3. There is **no** $|+\rangle \leftrightarrow |-\rangle$ entry. That transition is the microwave one and carries no cavity matrix element.

The asymmetry that eventually appears comes only from δ , through Eq. (6), and from which state the atoms are actually prepared in.

4 Step 3: many atoms

Take N atoms, all initially in $|g\rangle$, and work at low excitation,

$$n_+ + n_- \ll N. \quad (C2)$$

Holstein–Primakoff then replaces each collective excitation by a boson: $|g\rangle$ becomes the vacuum, and $b_\pm^\dagger$ creates one atom in $|\pm\rangle$, spread coherently over the ensemble. The couplings add constructively, so $\lambda \rightarrow \lambda\sqrt{N}$. Writing

$$G \equiv \lambda\sqrt{N} = \frac{g_0\Omega_c\sqrt{N}}{2\Delta}, \quad (9)$$

and keeping for now only the co-rotating terms that Eq. (8) supplies,

$$H = \omega_c \hat{a}^\dagger \hat{a} + \omega_+ b_+^\dagger b_+ + \omega_- b_-^\dagger b_- + \frac{G}{\sqrt{2}} (\hat{a}^\dagger b_+ + \hat{a}^\dagger b_- + \text{h.c.}). \quad (10)$$

Equation (10) conserves the total excitation number, so starting from $|g, 0\rangle$ nothing happens: it is an exact eigenstate. To obtain a Dicke model one must add counter-rotating terms $\hat{a}^\dagger b_\pm^\dagger + \text{h.c.}$, which is done by driving with two tones (or one modulated tone) so that a second Raman channel creates a cavity photon *and* excites the atom. With both present,

$$\boxed{H = \omega_c \hat{a}^\dagger \hat{a} + \sum_{s=\pm} \left[\omega_s b_s^\dagger b_s + \frac{G}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger) (b_s + b_s^\dagger) \right]} \quad (11)$$

— a Dicke model with *two* atomic modes.

5 Step 4: the threshold

Write Eq. (11) in quadratures $x_j = (o_j + o_j^\dagger)/\sqrt{2}$. The potential energy is $\frac{1}{2} \mathbf{x}^\top V \mathbf{x}$ with $\mathbf{x} = (x_a, x_+, x_-)$ and

$$V = \begin{pmatrix} \omega_c & \sqrt{2}G & \sqrt{2}G \\ \sqrt{2}G & \omega_+ & 0 \\ \sqrt{2}G & 0 & \omega_- \end{pmatrix}. \quad (12)$$

The normal phase (all atoms in $|g\rangle$, cavity dark) is stable while V is positive definite. It loses stability when $\det V = 0$, and using $\omega_+ + \omega_- = 2\delta$,

$$\det V = \omega_c \omega_+ \omega_- - 2G^2(\omega_+ + \omega_-) \implies \boxed{\omega_c \omega_+ \omega_- = 4G^2\delta} \quad (13)$$

This is the general result. Note that positive definiteness also requires $\omega_- > 0$: if the lower dressed state drops below $|g\rangle$, the normal phase fails for a mundane reason — a two-photon resonance — and not by symmetry breaking.

The two-level limit

Suppose the microwave splitting is large, so that one branch is far off resonance:

$$\Omega_\mu \gg \omega_-, \quad G, \quad \omega_c \quad \Longrightarrow \quad \omega_+ \simeq 2\delta \gg \omega_-. \quad (\text{C3})$$

Then $4G^2\delta \simeq 2G^2\omega_+$, and Eq. (13) collapses to

$$\boxed{\omega_c \omega_- = 2G^2} \quad \text{i.e.} \quad G_{\text{eff}} \equiv \frac{G}{\sqrt{2}}, \quad G_{\text{eff}}^c = \frac{1}{2}\sqrt{\omega_c \omega_-}, \quad (14)$$

which is exactly the textbook two-level Dicke threshold for a spin with splitting ω_- coupled at $G/\sqrt{2}$. *This* is the effective two-level model: the levels are $|g\rangle$ and the *near-resonant* dressed state, and the coupling carries a factor $1/\sqrt{2}$ from the dressing. Dropping the far branch without the $1/\sqrt{2}$, or building the spin on the far branch instead of the near one, both give wrong answers.

6 Summary of the reduction

Step	Condition	What it buys, and what breaks without it
Rotating frame + RWA	optical frequencies $\gg$ all Rabi rates	Eq. (1); ω_c, δ become tunable detunings, not optical frequencies
Eliminate $ 2\rangle$	$ \Delta \gg \Omega_c, g_0\sqrt{n}, \Gamma_e$	Raman coupling $\lambda = g_0\Omega_c/2\Delta$; leaves light shifts, one of which is the dominant systematic
Dress $ 3\rangle, 4\rangle$	microwave near resonance	$\omega_\pm = \delta \pm \Omega_\mu/2$: the measurand becomes a level position
Holstein–Primakoff	$n_+ + n_- \ll N$	collective enhancement $\lambda \rightarrow G = \lambda\sqrt{N}$
Add counter-rotating	second drive tone	without it, excitation number is conserved and $ g, 0\rangle$ never evolves
Drop the far branch	$\Omega_\mu \gg \omega_-, G, \omega_c$	the two-level Dicke model, coupling $G/\sqrt{2}$, spin splitting ω_-

The chain of reasoning is: four levels $\rightarrow$ (eliminate $|e\rangle$) $\rightarrow$ three levels $\rightarrow$ (dress) $\rightarrow |g\rangle$ plus two branches $\rightarrow$ (many atoms, counter-rotating terms) $\rightarrow$ two-branch Dicke $\rightarrow$ (one branch far detuned) $\rightarrow$ two-level Dicke. The last arrow is the only one that is optional, and it is the one most easily got wrong.